

John M. Ortega

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EDUCATION

CUNY: City College

New York, NY

Bachelor of Science in Computer Science, Minor in Mathematics | **GPA: 3.8/4.0**

December 2027

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Calculus 1-3, Software Engineering

2x Hackathon Winner: Morgan Stanley Code To Give 2026, Brooklyn Hacks 2026

EXPERIENCE

AI Engineer Intern

Jun 2026 – Present

U.S. Air Force Research Laboratory (AFRL)

- Built vector, vectorless, and knowledge graph (KG) RAG pipelines using local ollama models and LoRA fine-tuning.
- Achieved **95%** KG rescue rate when Vectorless pipeline missed docs by adding KG fallback+merge hybrid path.
- Built an admin pipeline turning preference judgments into curated training data for offline LoRA fine tuning.

Software Engineer Intern

Feb 2026 – May 2026

City University of New York (CUNY)

New York, NY

- Integrated **Next.js** and **React** with **Django** REST API endpoints for delivery of generated academic plans.
- Built Django REST API endpoints to generate personalized multi-semester academic plans across **500+** courses.
- Engineered prerequisite dependency & constraint-based scheduling algorithms, reducing course enrollment time by **77%**
- Designed a fitness function for optimal course selection, reducing invalid graduation paths by **99%**.

Software Engineer Intern

Jul 2025 – Sep 2025

The Difference App LLC

New York, NY

- Developed full-stack admin console using **Node.js**, **Express.js**, and **MySQL** to manage **1,500+** user fitness records.
- Reduced **20%** of backend codebase by refactoring routes into reusable functions, eliminating **80+** lines of code.
- Improved login security by resolving a token handling vulnerability, reducing unauthorized login attempts by **99%**.

PROJECTS

Morgan Stanley Hackathon – 1st Place | *Next.js, PostgreSQL, React, Google Maps API, Gemini API* [Link](#)

- Built a self-serve flyer platform for Lemontree, enabling independent print and distribution campaign management.
- Designed PostgreSQL schemas with indexes, triggers, and constraints for volunteer coordination and route tracking.
- Built a printer locator using the Google Maps Places API with Overpass API fallback for broader coverage.
- Implemented a real-time leaderboard with multi-metric scoring, time-filtered rankings, and automated certificates.

Royal Flush | *React, Node.js, Express, PostgreSQL, Docker, GCP*

[Link](#) | [GitHub](#)

- Shipped production app to **GCP** by Dockerizing React/Node.js across **Cloud Run, Cloud SQL, and GCS**.
- Secured DB/JWT secrets in production, by configuring **IAM, Secret Manager**, and GCS signed URLs.
- Hardened API security by implementing **JWT** auth, **BCrypt** hashing, and rate limiting on all **Express** endpoints.
- Supported location-based reviews & favorites, by designing a **5-table PostgreSQL** schema synced to Google Place IDs
- Resolved JWT authentication failures in production by implementing a same-origin **NGINX** reverse proxy.

RAG Study Assistant Discord Bot | *Python, LangChain, Pinecone, ChromaDB, Discord.py*

[Github](#)

- Built a RAG Discord bot to automate study material retrieval from **500+** course PDFs, eliminating manual searching.
- Scaled deployment across **50+** servers, successfully processing **300+** queries and supporting **200+** students.
- Built a LangChain/OpenAI pipeline to chunk PDFs into semantic embeddings and vector representations.
- Enabled real-time ingestion for **500+** users, simplifying the transformation of study materials into searchable data.

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, C/C++, SQL, HTML, TypeScript

Frameworks & Libraries: Django, React, Node.js + Express.js, Spring Boot, Next.js

Tools: MongoDB, Supabase, MySQL, PostgreSQL, Firebase, GCP, Docker, Cursor

AI/ML: PyTorch, NumPy, Matplotlib, Seaborn, LangChain, Claude